

Ans 1 As we know that g_m (Transconductance) = $\frac{\partial I_p}{\partial V_{GS}} \approx \frac{\Delta I_p}{\Delta V_{GS}}$

Now at $V_p = 4750V$
(Plate Voltage)

$$g_m = \frac{1.4 - 1}{25 - 0} = 16 \text{ mmo}$$

$$\text{Now } r_o = \frac{\partial V_{PS}}{\partial I_p} \approx \frac{\Delta V_{PS}}{\Delta I_p}$$

$$[V_{PS} = V_{\text{plate}} - V_{\text{screen}}]$$

$$\Rightarrow \text{At } V_G \approx 75V$$

$$r_o = \frac{6000 - 3250}{2.6 - 2.5} = 6875 \Omega$$

Ans 2 (a) For NMOS, Observing the graph we get $V_{GS} = I_d = 1.3 \text{ mA}$
at $V_{GS} = 0.6V$ and $V_{DS} = 0.6V$

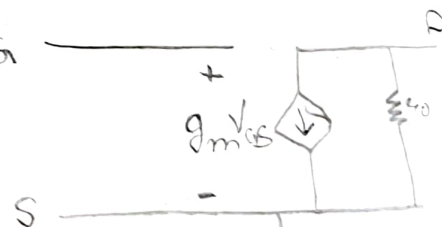
$$\Rightarrow g_m = \frac{\partial I_D}{\partial V_{GS}} \approx \frac{I_D}{V_{GS}} = \frac{1.3}{0.6} \times 10^{-3} \approx 2.17 \text{ mmo}$$

• In small signal model; $V_{DS} = -g_m V_{GS} r_o$

$$r_o = 1/g_m$$

$$(\because V_{GS} = V_{DS} = 0.6V)$$

$$= 1/2.17 \times 10^{-3} \approx 460.83 \Omega$$



→ For PMOS, From graph we get $I_d = -0.9 \times 10^{-3} A$
at $V_{GS} = -0.6V$ and $V_{DS} = -0.6V$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \approx \frac{I_D}{V_{GS}} = \frac{-0.9 \times 10^{-3}}{-0.6} = 1.5 \text{ mmo}$$

$$r_{o1}, r_{o2} = \frac{1}{g_m} = \frac{1}{1.5 \times 10^{-3}} = 666.67 \Omega$$

⑥ For NMOS —: At $V_{DS} = 0.6V$

$$\rightarrow \text{For } V_{GS1} = 0.6V, I_{D1} = 1.3mA$$

$$\rightarrow \text{For } V_{GS2} = 0.5V, I_{D2} = 0.75mA$$

Taking saturation into account

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2$$

$$\Rightarrow \frac{1.3}{0.75} = \left(\frac{0.6 - V_T}{0.5 - V_T} \right)^2$$

Solving above we get $V_T = 0.185V$

For PMOS —: At $V_{DS} = -0.6V$

$$\rightarrow \text{For } V_{GS1} = -0.6V, I_{D1} = 0.9mA$$

$$\rightarrow \text{For } V_{GS2} = -0.5V, I_{D2} = 0.5mA$$

Taking saturation into account

$$I_D = \frac{0.9}{0.5} = \left(\frac{0.6 - |V_T|}{0.5 - |V_T|} \right)^2$$

$$\Rightarrow V_T = -0.208V$$

From observation we can say that the curve is quadratic.

⑦ For NMOS —:

Using straight line approximation (At $V_{GS} = 0.6V$)

$$I_D = m V_{DS} + \text{constant where}$$

$$m = \frac{1.16 - 1.04}{0.3 - 0.2} \times 10^{-3} = 1.2 \times 10^{-3} \text{ mho}$$

Now,

$$1.04 \times 10^{-3} = 1.2 \times 10^{-3} \times 0.2 + C$$

$$\Rightarrow C = 0.8 \times 10^{-3}$$

$$\Rightarrow V_A = \frac{-C}{m} = \frac{-0.8 \times 10^{-3}}{1.2 \times 10^{-3}} = -0.667 \text{ V}$$

$$m = \frac{0.72 - 0.69}{0.2 - 0.3} \times 10^{-3} = -0.8 \times 10^{-3}$$

Now, $0.72 = -0.8 \times 10^{-3} \times (-0.3) + C$

$$\Rightarrow C = 0.48 \times 10^{-3}$$

So, $V_A = \frac{-C}{m} = \frac{-0.48 \times 10^{-3}}{0.8 \times 10^{-3}} = 0.6 \text{ V}$

Ans 3 # For NMOS,

While observing I_D vs V_{GS} graph we got that $V_T \approx 0.4 \text{ V}$

Now from I_D vs V_{DS} graph for different V_{GS}

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \frac{654 - 564}{0.2} \times 10^{-6} = 450 \mu \text{ mho}$$

$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1.2 - 1}{(664 - 654) \times 10^{-6}} \approx 20 \text{ k}\Omega$$

After using straight line Approximation

$$I_D = mV_{DS} + C$$

$$m = \frac{\partial I_D}{\partial V_{DS}} = \frac{50}{10} \times 10^{-6}$$

So, $654 \times 10^{-6} = \frac{50}{10} \times 10^{-6} \times 1 + C$

$$\Rightarrow C = 604 \times 10^{-6}$$

$$\Rightarrow V_A = \frac{-C}{m} = \frac{-604 \times 10^{-6}}{49.50 \times 10^{-6}} = -12.1 \text{ V}$$

For PMOS

from I_d vs V_{GS} graph

$$V_T \approx -4.15 \text{ Volt}$$

From I_D vs V_{DS} graph

$$\text{At } V_{DS} = -6 \text{ Volt } g_m = \frac{\partial I_D}{\partial V_{GS}} \approx 0.405 \text{ mho}$$

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